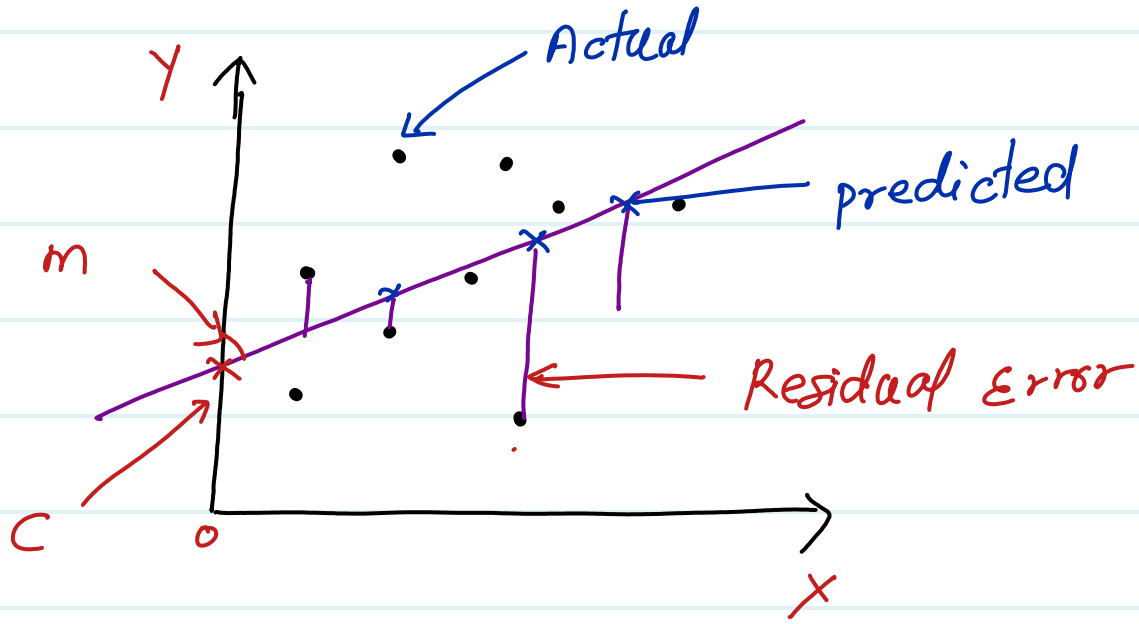
Linear Regression



Line eqn –

$y = mx + c$

y = Actual data

$\hat{y}$ = Predicted data

$$y = m_1x_1 + m_2x_2 + m_3x_3 + \cdots + m_nx_n + c$$

y = Dependent variable

x = Independent variable

m = slop

c = Intercept

$y - \hat{y}$ = Residual error

Base equation

$$y = mx + c$$

OR

$$y = h_\theta(x)$$

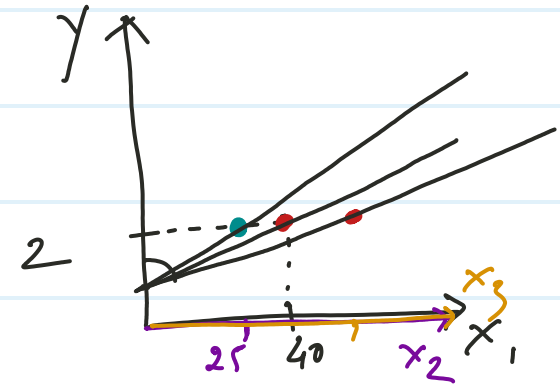
OR

Simple linear eqn. $h_\theta(x) = \theta_0 + \theta_1 x_1$

multi linear eqn $h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4$

$$y = J = h_\theta(x)$$

$$J = J(\theta_0, \theta_1)$$



Loss function

$$J(\theta_0, \theta_1) = (y - \hat{y})^2$$

It only calculate single datapoint

cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left[h_\theta(x)^{(i)} - y^{(i)} \right]^2$$

This is cost function to min. error by changing value of θ_0, θ_1

θ_0 = slop

θ_1 = intercept

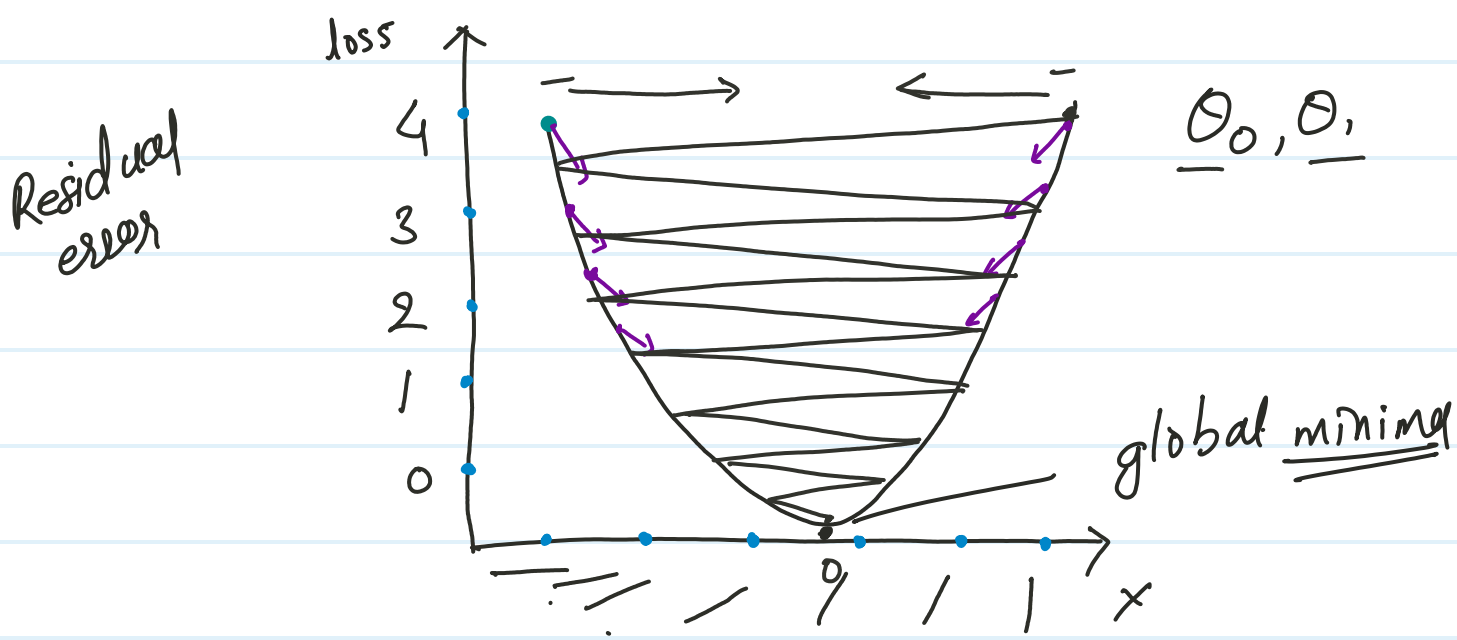
$\theta_0 = 40$
$\theta_1 = 2$

$\theta_0 = 45$
$\theta_1 = 2.5$

Repeat conversion therom

$$\theta_j = \theta_j - \alpha \frac{\partial J(\theta)}{\partial \theta_j}$$

gradient decent graph



$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left[h_\theta(x)^i - y^i \right]^2$$

$$\boxed{\theta_j = \theta_j - \alpha \frac{\partial J(\theta)}{\partial \theta_j}}$$ update eqn^n

for θ_0

$$\frac{\partial J\theta}{\partial \theta_0} = \frac{1}{m} \sum_{i=1}^{m} (\overset{\hat{y}}{h_\theta(x)^i} - y^i) \quad \text{— (1)}$$

for θ_1

$$\frac{\partial J\theta}{\partial \theta_1} = \frac{1}{m} \sum_{i=1}^{m} (\overset{\hat{y}}{h_\theta(x)^i} - y^i) * x^i \quad \text{— (2)}$$

update $\theta_0 = \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x)^i - y^i)$

update $\theta_1 = \theta_1 - \alpha \frac{1}{m} \sum (h_\theta(x)^i - y^i) * x^i$

α is learning rate

(0.01, 0.05, 0.1, 0.25, 0.3, 0.4 . - - - -)

(θ_0, θ_1)

* Model evaluation or performance metrix

(i) MSE (mean squared error)
(ii) RMSE (Root mean square error)
(iii) MAE (mean Absolute error)
(iv) R^2
(v) Adj. R^2

2 - 3	(-1) - 1
3 - 4	(-1) - 1
5 - 5	(0) - 0

(i) MSE

$$MSE = \sum_{i=1}^{n} \frac{(y - \hat{y})^2}{n}$$

$\frac{2}{3} \Rightarrow 0.5$

② RMSE

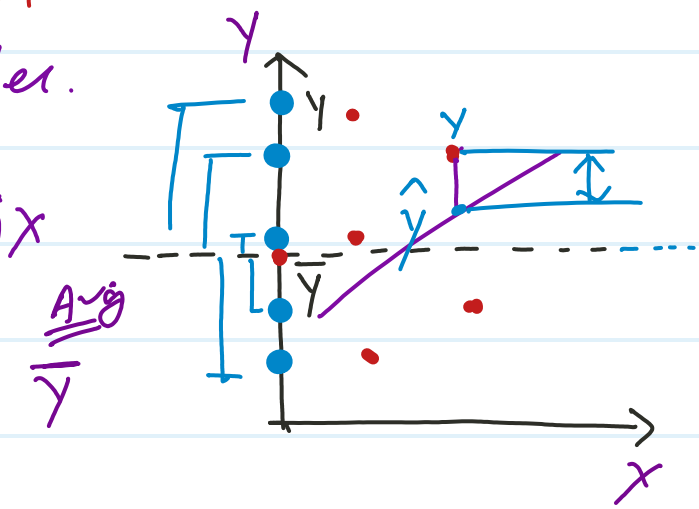
$$RMSE = \frac{1}{n}\sqrt{\sum_{i=1}^{n}(y-\hat{y})^2}$$

③ MAE

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y-\hat{y}|$$

lower value better.

* Accuracy Matrix



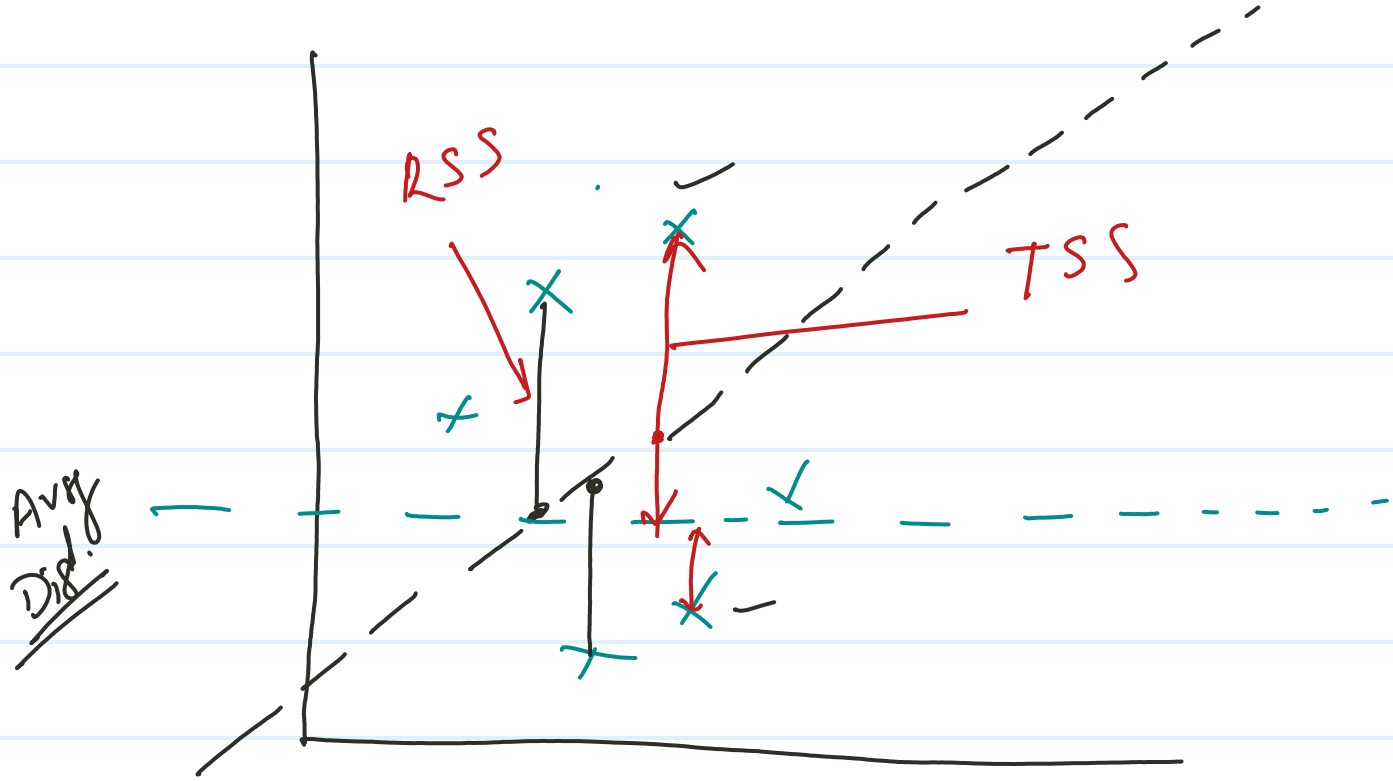
$$R^2 = 1 - \frac{RSS}{TSS}$$

R^2 = coeff. of determination

RSS = sum of square of residual

RSS = Distance b/w y and $\hat{y}$

TSS = Distance b/w y and $\bar{y}$



$$RSS = \sum (y - \hat{y})^2$$

$$TSS = \sum (y - \bar{y})^2$$

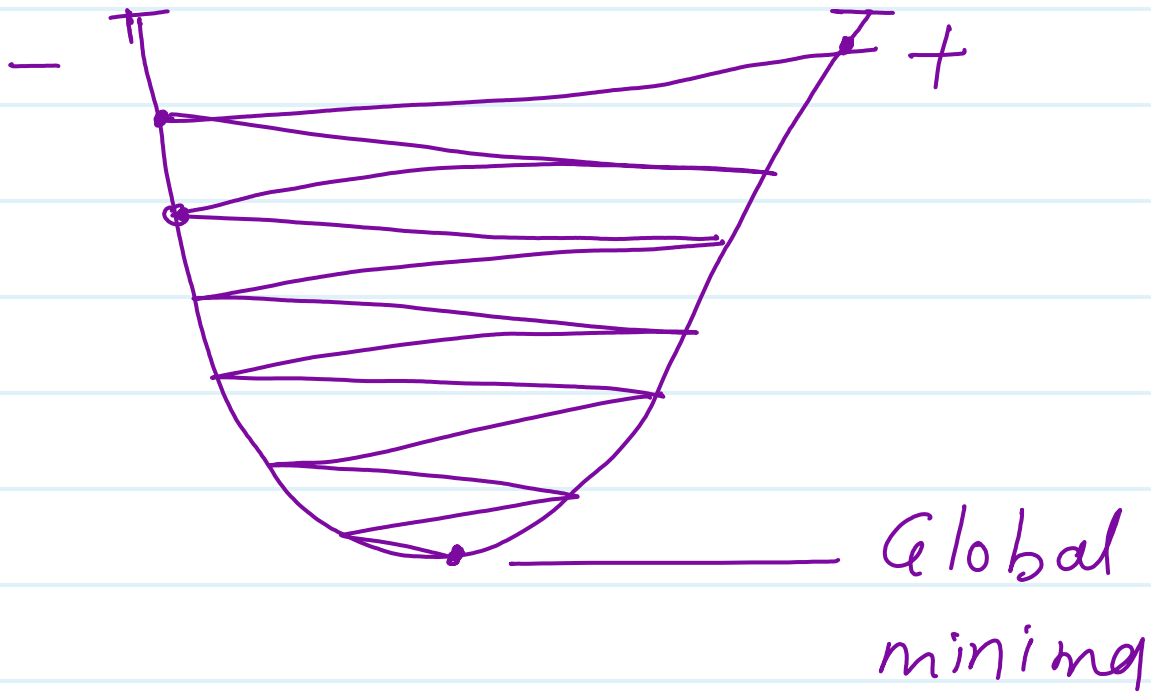
TSS Avg distance

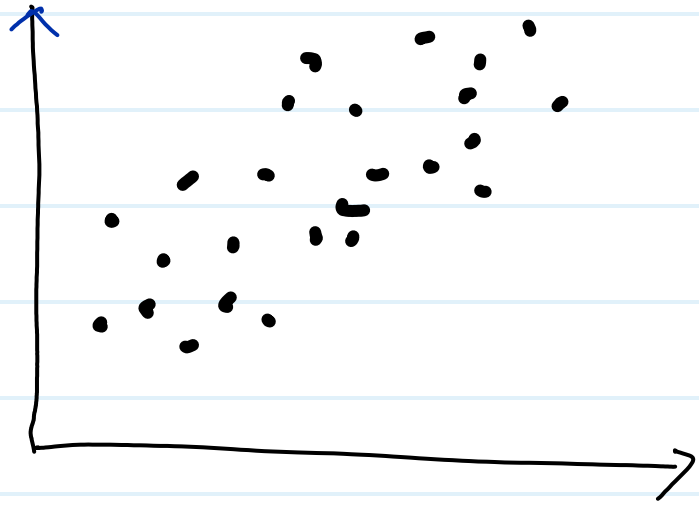
② Adj. R^2

$$\text{Adj. } R^2 = 1 - \frac{(1 - R^2)(N - 1)}{N - P - 1}$$

N = no. of datapoint in our dataset

P = no. of independ variable (x_1 x_2 x_3 ----)





* To find multi co-linearity

X_1	X_2	X_3	X_4	Y
←			→	↔

$X_1 X_2$ $X_2 X_3$

$X_1 X_3$ $X_2 X_4$

$X_1 X_4$ $X_3 X_4$

$\boxed{X_1 X_3 X_4 \; Y}$

*

VIF (variance inflation factors)

$$VIF = \frac{1}{1-R^2}$$

X_1 - 3
X_2 - 2
X_3 - 4
X_4 - 6 ✗

VIF = start 1 and it has no limit

IF 1 or less than 5 so no. multicolinearity

If > 5 so there will be co-linearity b/w inde. Feature.

over fitting :-

low biased
high variance

under fitting

high biased
low variance

Best fitting

low biased
low varianced

$$h_\theta = \theta_0 + \theta_1 x_1$$

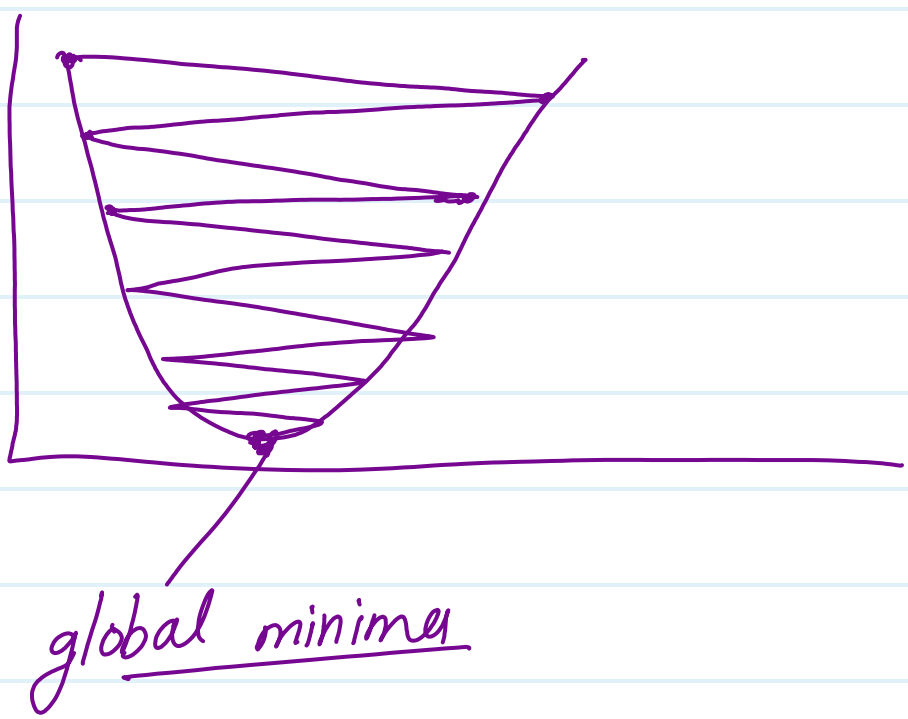
$$h_\theta = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \text{-----} + \theta_n x_n$$

$\theta_1 = 2$ cost $= 0.5$

$$\boxed{\theta_{1\,new} = \theta_{1\,old} - \eta \frac{\partial \theta_{loss}}{\partial \theta_{1\,old}}}$$

$\theta_{1\,new} =$

$$\theta_{0\,ne} = \theta_{0\,new} - \eta \frac{\partial \theta_{loss}}{\partial \theta_{0\,old}}$$



Rooms	sqe	price
2	1000	30L
3	1100	35L
2	700	25L
2	800	28-1. ?

$$= 1 - \frac{RSS}{TSS} \qquad = 0.40 \qquad = 0.20$$

$$= 1 - \frac{0.40}{0.20}$$

$$= 1 - 0.2 \Rightarrow 0.8 \quad 80\%$$

$$\Rightarrow 1 - \frac{0.20}{0.40}$$

$$\Rightarrow 1 - 0.05 \Rightarrow 0.95 = 95\%$$

Assumptions - multicolinearity
homoscedacity